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Experimental Dialog Correlation and Recovery Across Agent Adapters

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Problem Description

Multi-agent AI systems (e.g., using MCP and A2A) manage conversation context in framework-specific ways. When an agent restarts or a network link drops mid-task, there is no standard way to identify the task, its state, or whether a request was already executed, leading to lost context and duplicated side effects.

Framework-specific context management

Different agent frameworks (MCP sessions, A2A task and context IDs) use their own identifiers and formats, with no common mechanism across adapters.

Context loss on failures

When an agent restarts or a network link drops mid-task, the conversation context and task state are lost, making it unclear which message belongs to which task or what state it was in.

Risk of duplicated side effects

Retries after failures can execute the same request multiple times (e.g., provisioning the same network slice twice), which is costly and unsafe in network operations.

No standard, cross-adapter solution

There is currently no standard solution to correlate requests, validate lifecycle states, deduplicate retries, and recover tasks durably across heterogeneous agent frameworks.



Background

Multi-agent AI systems use different frameworks such as MCP and A2A, each maintaining conversation context and task identity in their own way. Network disconnects or agent restarts can interrupt a task and leave the state uncertain, making it hard to recover safely. This is critical in 6G network orchestration, where duplicated actions (e.g., provisioning a network slice twice) can waste resources.



1 **Different Frameworks**

Agents use frameworks like MCP and A2A with different context and task management mechanisms.

2 **Context is Framework-specific**

MCP and A2A maintain conversation context and task identity in different formats.

3 **Failures Interrupt Tasks**

A network disconnect or agent restart can leave the task state uncertain and context lost.

4 **Need for Continuity**

Safe recovery requires persistent identifiers, lifecycle tracking, request deduplication, and durable storage.

5 **Critical for 6G**

In network orchestration, duplicated actions such as provisioning a network slice twice can waste resources and cause operational issues.

Literature & RFC Review

We reviewed existing frameworks, proposals and standards related to agent communication, context management, reliability and idempotency to identify relevant ideas and gaps addressed by DialogSync.

Model Context Protocol (MCP)

Framework for tool and context integration

- Defines sessions and carries context via `_meta´ extension point.
- Provides a standardized way to attach metadata to messages and tool calls.
- Maintains conversation context in a framework-specific format (e.g., session ID).
- Inspires our design for embedding `dialog_id´, `task_id´ and state metadata.

Agent2Agent (A2A)

Framework for agent communication

- Uses `taskId´ and `contextId´ to identify tasks and maintain conversation context.
- Defines message shapes and metadata for cross-agent communication.
- Supports task lifecycle operations (e.g., task, cancel).
- Guides our approach for carrying DialogSync schema in A2A metadata.

SQLite Write-Ahead Logging (WAL)

Durable local state and crash recovery

- Provides durable, crash-safe writes with atomic commit.
- Ensures recovery after unexpected restarts or crashes.
- Suitable for lightweight, embedded prototypes without external dependencies.
- Used in DialogSync for write-ahead outbox and effect-intent journal.

IETF agentproto working group

Standardization efforts for agent communication

- Discusses protocols, message formats and interoperability for agent systems.
- Covers ideas around context exchange, capability discovery and agent interaction patterns.
- Provides useful problem space and design considerations.
- Our schema `dsync-exp/0.1´ is experimental and not an IETF draft or standard.

Key Gap Addressed

While MCP and A2A provide framework-specific context and task identifiers, there is no common, cross-adapter mechanism for lifecycle validation, request deduplication and durable recovery — which is the gap addressed by DialogSync.

Proposed Solution

A framework-neutral layer for reliable task continuity across agent adapters.

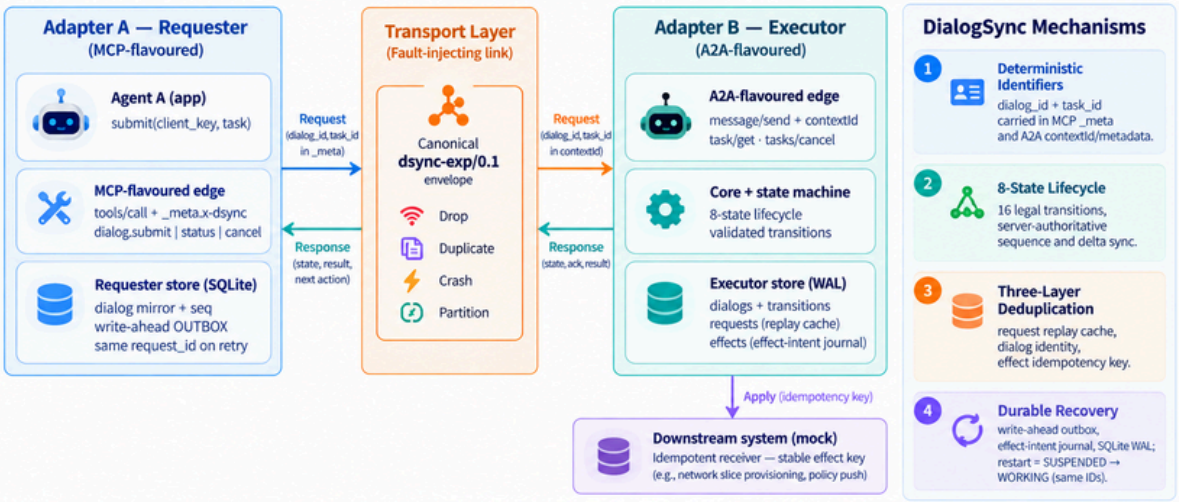


Optimization Proposed

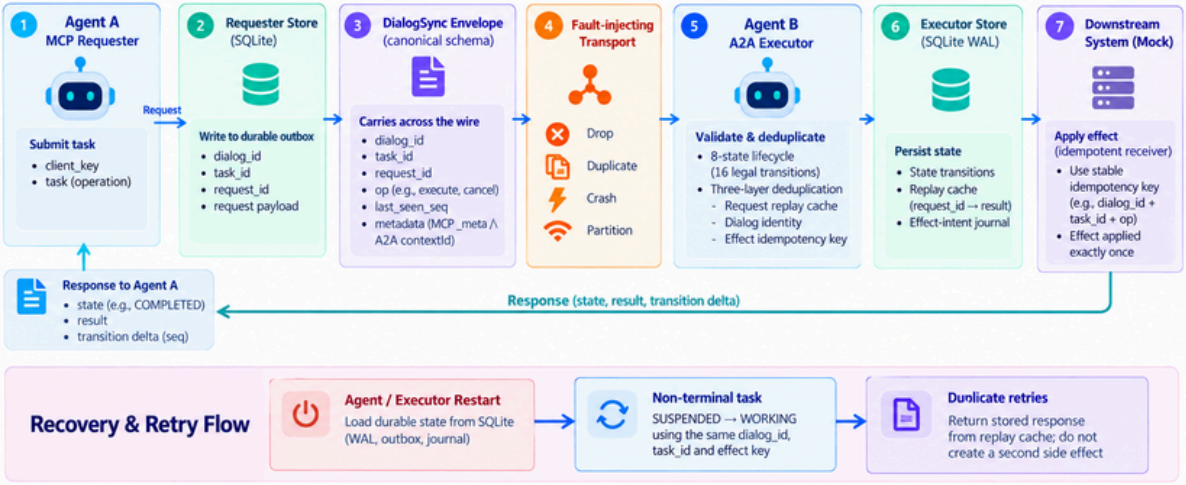
Key optimizations and enhancements introduced by DialogSync to ensure reliable task continuity across agent adapters.

Aspect	Before (Without DialogSync)	After (With DialogSync)
Task correlation	Framework-specific IDs (e.g., MCP session ID, A2A contextID); context can be lost during restarts or link drops.	**Stable dialog_id and task_id** carried across adapters and persist through restarts.
Lifecycle	Unclear or inconsistent task state across frameworks.	**8-state lifecycle with 16 legal transitions**, enforced on both adapters, with server-authoritative sequence and delta sync.
Retries	Duplicate side effects possible on retries (lost acknowledgements, replayed requests, etc.).	**Three-layer deduplication** (request replay cache, dialog identity, effect idempotency key) with rejection of conflicting request_id reuse.
Recovery	Agent restart may orphan tasks or repeat work, leading to inconsistent state and duplicate actions.	**SQLite WAL, write-ahead outbox and effect-intent journal**, with recovery through SUSPENDED → WORKING using the same IDs.
Measured outcome	Naive baseline creates **2–8 duplicate side effects** in tested fault scenarios (e.g., executor crash, lost ACK, retry storm).	**One side effect per task in all tested scenarios** (8/8 runs pass, 5 scenarios, 4 crash points), with an idempotent downstream receiver assumption.

Solution Architecture & Design

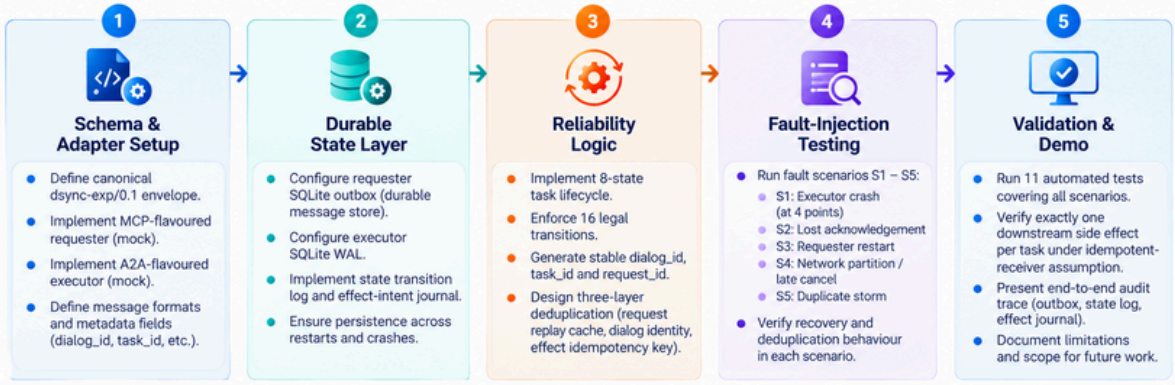


Data Flow Diagram



Deployment Timeline

A phased plan to build, test and deploy DialogSync with reliability and idempotency across agent adapters.



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Thank You for Your Attention!